Gabriel **Pedde Ungureanu** | Physics PhD | Quantum Engineer

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Profile

Theoretical physics PhD (6 papers, ∼60 citations) spanning quantum field theory and quantum information: a Physical Review D publication on **quantum complexity** (28 citations), hands-on **gate-based (Qiskit, Cirq), annealing (D-Wave) and neutral-atom (Pulser)** platforms, and HPC-scale quantum simulation (PETSc/SLEPc, Quantum ESPRESSO).

Research & Engineering Experience

SISSA — **Trieste, Italy**
Doctoral Researcher in Theoretical Particle Physics — *2022–2026*

- Exact, non-perturbative results in supersymmetric gauge theories and dualities via large-scale symbolic computation on Ulysses (SISSA EuroHPC); **6 published papers, ∼60 citations**.
- Published on **holographic quantum complexity** of de Sitter bubbles (Phys. Rev. D, **28 citations**), at the interface of quantum information and gravity.
- Building an LLM **"AI co-scientist"** for physics discovery, refereed by a deterministic symbolic/numerical evaluator; recovers exact General Relativity solutions to **machine precision**.

Education

SISSA / ICTP Joint Programme — **Trieste, Italy**
Master in High Performance Computing and AI (MHPC) — *2025–2026*
Concurrent with the last year of PhD. Quantum computing (CINECA QC-Lab: gate-based, annealing, neutral atoms, HPC–QC integration), numerical eigensolvers, machine learning, distributed GPU computing.

Università Cattolica del Sacro Cuore — **Brescia, Italy**
BSc Mathematics & MSc Physics — *2017–2022*
Theoretical physics, QFT, condensed matter. MSc thesis in **quantum complexity theory** at the interface of holography and quantum information.

Awards & Publications

6 peer-reviewed papers in theoretical physics (∼**60 citations**) — INSPIRE-HEP | Google Scholar

Best graduate of the year, MSc Physics — Università Cattolica del Sacro Cuore

110/110 *cum laude* — MSc Physics | 110/110 *cum laude* — BSc Mathematics

Istituto Toniolo merit fellowship — BSc, Università Cattolica del Sacro Cuore

Selected Projects — Quantum Computing & Simulation

Multi-platform QC: Hands-on quantum computing across three hardware paradigms (CINECA QC-Lab, Leonardo Quantum Suite): **Grover search in Cirq** and quantum algorithms in **Qiskit**; combinatorial optimisation as **QUBO** on **D-Wave quantum annealers** (constraint encoding, budget/placement problems); Maximum Independent Set via **QAA and QAOA** pulse sequences on Pasqal's **neutral-atom** platform (Pulser).

Schrödinger solver: Parallel real-space eigensolver for the time-independent Schrödinger equation in **2D/3D** (PETSc DMDA + **SLEPc**, MPI-distributed finite differences): hydrogen spectrum to **<1 % accuracy** on excited states, Kronig–Penney **band-gap opening** validated against nearly-free-electron theory; VTK/ParaView visualisation | GitHub

Electronic structure: Strong-scaling study of **Quantum ESPRESSO's Davidson iterative eigensolver** on Leonardo (CINECA): plane-wave MPI parallelism, ScaLAPACK distributed diagonalisation and **k-point pools** (**3× speedup** over 50 k-points), plus a **CUDA-offload build** (nvfortran); FCC band-structure calculations with FFT-based operators. |GitHub

Technical Skills

Quantum: Qiskit, Cirq, D-Wave Ocean (QUBO/annealing), Pulser (neutral atoms), QAOA; quantum information & complexity, QFT, band theory

Simulation: PETSc/SLEPc, Quantum ESPRESSO, Davidson & Krylov eigensolvers, finite differences, FFT methods

Programming: Python, C++, C, Fortran, Mathematica, LaTeX

Scale / HPC: CUDA, cuBLAS, OpenACC, MPI, OpenMP, SLURM; Leonardo (CINECA), Ulysses (SISSA), Argo (ICTP)

AI / ML: PyTorch, Hugging Face, reinforcement learning (PPO/RLOO), LLM agents (LangGraph, RAG)

Tools: Git, Docker, Singularity/Apptainer, HDF5/NetCDF

Languages: Italian (native) | English (C1) | Romanian (A2)

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